

Engineering Mathematics: System of Linear Equations

1. Introduction to Matrix Representation

A system of linear equations involves multiple equations with shared variables. To solve these efficiently, we translate the algebraic form into a matrix equation.

Algebraic Form

$$a_1x + b_1y + c_1z = d_1$$

$$a_2x + b_2y + c_2z = d_2$$

$$a_3x + b_3y + c_3z = d_3$$

Matrix Form ($AX = B$)

The system is represented as:

$$\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

- A (Coefficient Matrix):** Contains the coefficients of the variables.
- X (Variable Matrix):** A column matrix of the unknowns (x, y, z).
- B (Constant Matrix):** A column matrix of the constants on the right-hand side.

2. Classification of Systems

Based on the constant matrix (B), systems are classified into two types:

- Non-Homogeneous System:** $B \neq 0$ (at least one constant is non-zero).
- Homogeneous System:** $B = 0$ (all constants are zero, represented as $AX = 0$).

3. The Augmented Matrix $[A|B]$

To analyze and solve the system, we combine Matrix A and Matrix B into a single structure called the **Augmented Matrix**. This allows us to perform row operations on the entire system simultaneously.

$$[A|B] = \left[\begin{array}{ccc|c} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \end{array} \right]$$

4. Consistency and Types of Solutions

A system is **Consistent** if at least one solution exists, and **Inconsistent** if no solution exists.

Condition (for 2 variables)	Algebraic Meaning	Solution Type
$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$	Lines intersect at one point	Unique Solution
$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$	Lines are parallel	No Solution
$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$	Lines are coincident (overlap)	Infinite Solutions

5. Step-by-Step Transformation Example

We use Elementary Row Operations (ERO) to transform the Augmented Matrix into Row Echelon Form to find the solution.

Goal: Solve the system by creating zeros in the lower triangle.

Initial Augmented Matrix

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 14 \\ 0 & 1 & 2 & 8 \\ 2 & 4 & 7 & 31 \end{array} \right] \xrightarrow{(R_3 \rightarrow R_3 - 2R_1)} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 14 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

Back-Substitution Logic

Once in Echelon form, we solve from the bottom up:

- 1. From R_3 : $1z = 3 \implies z = 3$
- 2. From R_2 : $y + 2z = 8 \implies y + 2(3) = 8 \implies y = 2$
- 3. From R_1 : $x + 2y + 3z = 14 \implies x + 2(2) + 3(3) = 14 \implies x = 1$

Solution: $(x, y, z) = (1, 2, 3)$

Working Rule for Consistency of Linear Equations

1. Represent the System:

Let the given system of linear equations be represented in matrix form as:

$$AX = B \quad \dots (1)$$

2. Form the Augmented Matrix:

Find the augmented matrix $[A|B]$ for the system (1).

3. Row Reduction:

Transform the augmented matrix $[A|B]$ into **Echelon Form**. Once in echelon form, observe the rank of the coefficient matrix A , denoted as $\rho(A)$, and the rank of the augmented matrix $[A|B]$, denoted as $\rho(A|B)$.

4. Consistency Analysis:

- **Case 1: Inconsistent System**
If $\text{Rank}(A) \neq \text{Rank}(A|B)$, the system is **inconsistent** and has **no solution**.
- **Case 2: Consistent with Unique Solution**
If $\text{Rank}(A) = \text{Rank}(A|B) = n$ (where n is the number of unknown variables), the system is **consistent** and has a **unique solution**.
- **Case 3: Consistent with Infinite Solutions**
If $\text{Rank}(A) = \text{Rank}(A|B) = r < n$ (where n is the number of variables), the system is **consistent** and has **infinitely many solutions**.
 - In this case, $(n - r)$ variables must be chosen arbitrarily.